

1.

```
mysql> SELECT `customer-name`  
        -> FROM borrower_relation  
        -> UNION  
        -> SELECT `customer-name`  
        -> FROM depositor_relation;
```

customer-name
Adams
Curry
Hayes
Jackson
Jones
Smith
Williams
Johnson
Lindsay
Turner

10 rows in set (0.00 sec)

2.

```
mysql> SELECT*  
        -> FROM borrower_relation as a  
        -> inner join depositor_relation as b  
        -> ON a.`customer-name`=b.`customer-name`;
```

customer-name	loan-number	customer-name	account_number
Hayes	L-15	Hayes	A-102
Jones	L-17	Jones	A-217
Smith	L-11	Smith	A-215
Smith	L-23	Smith	A-215

4 rows in set (0.00 sec)

3.

```
mysql> SELECT*
-> FROM
-> (SELECT*
-> FROM loan_relation as a
-> inner join borrower_relation as b
-> ON a.loan_number=b.'loan-number'
-> ) as c
-> left join
-> (SELECT*
-> FROM account_relation as a
-> inner join depositor_relation as b
-> ON a.accountnumber=b.account_number
-> ) as d
-> ON c.'customer-name'=d.'customer-name'
-> WHERE d.'customer-name' IS NULL;
```

loan_number	branch_name	amount	customer-name	loan-number	accountnumber	branch_name	balance	customer-name	account_number
L-16	Perryridge	1300	Adams	L-16	NULL	NULL	NULL	NULL	NULL
L-93	Mianus	500	Curry	L-93	NULL	NULL	NULL	NULL	NULL
L-14	Downtown	1500	Jackson	L-14	NULL	NULL	NULL	NULL	NULL
L-17	Downtown	1000	Williams	L-17	NULL	NULL	NULL	NULL	NULL

4 rows in set (0.00 sec)

4.

```
mysql> SELECT DISTINCT b1.branchname
-> FROM branch_relation b1
-> JOIN branch_relation b2 ON b1.assets > b2.assets
-> WHERE b2.branch_city = 'Brooklyn';
```

branchname
Downtown
Round Hill

2 rows in set (0.00 sec)

5.

```
mysql> SELECT branchname
-> FROM branch_relation
-> WHERE assets > ANY (
->     SELECT assets
->     FROM branch_relation
->     WHERE branch_city = 'Brooklyn'
-> );
```

branchname
Downtown
Round Hill

```
2 rows in set (0.00 sec)
```

6.

```
mysql> SELECT branch_name
-> FROM account_relation
-> GROUP BY branch_name
-> ORDER BY AVG(balance) DESC
-> LIMIT 1;
```

branch_name
Brighton

```
1 row in set (0.01 sec)
```

```
mysql> |
```

7.

```
mysql> SELECT `customer-name`  
-> FROM borrower_relation b  
-> WHERE EXISTS (  
->     SELECT 1  
->     FROM depositor_relation d  
->     WHERE b.`customer-name` = d.`customer-name`  
-> );  
+-----+  
| customer-name |  
+-----+  
| Hayes         |  
| Jones         |  
| Smith         |  
| Smith         |  
+-----+  
4 rows in set (0.00 sec)
```

8.

```
mysql> SELECT*
-> FROM loan_relation
-> NATURAL JOIN borrower_relation;
```

loan_number	branch_name	amount	customer-name	loan-number
L-93	Mianus	500	Adams	L-16
L-23	Redwood	2000	Adams	L-16
L-17	Downtown	1000	Adams	L-16
L-16	Perryridge	1300	Adams	L-16
L-15	Perryridge	1500	Adams	L-16
L-14	Downtown	1500	Adams	L-16
L-11	Round Hill	900	Adams	L-16
L-93	Mianus	500	Curry	L-93
L-23	Redwood	2000	Curry	L-93
L-17	Downtown	1000	Curry	L-93
L-16	Perryridge	1300	Curry	L-93
L-15	Perryridge	1500	Curry	L-93
L-14	Downtown	1500	Curry	L-93
L-11	Round Hill	900	Curry	L-93
L-93	Mianus	500	Hayes	L-15
L-23	Redwood	2000	Hayes	L-15
L-17	Downtown	1000	Hayes	L-15
L-16	Perryridge	1300	Hayes	L-15
L-15	Perryridge	1500	Hayes	L-15
L-14	Downtown	1500	Hayes	L-15
L-11	Round Hill	900	Hayes	L-15
L-93	Mianus	500	Jackson	L-14
L-23	Redwood	2000	Jackson	L-14
L-17	Downtown	1000	Jackson	L-14
L-16	Perryridge	1300	Jackson	L-14
L-15	Perryridge	1500	Jackson	L-14
L-14	Downtown	1500	Jackson	L-14
L-11	Round Hill	900	Jackson	L-14
L-93	Mianus	500	Jones	L-17
L-23	Redwood	2000	Jones	L-17
L-17	Downtown	1000	Jones	L-17
L-16	Perryridge	1300	Jones	L-17
L-15	Perryridge	1500	Jones	L-17

L-16	Perryridge	1300	Jones	L-17
L-15	Perryridge	1500	Jones	L-17
L-14	Downtown	1500	Jones	L-17
L-11	Round Hill	900	Jones	L-17
L-93	Mianus	500	Smith	L-11
L-23	Redwood	2000	Smith	L-11
L-17	Downtown	1000	Smith	L-11
L-16	Perryridge	1300	Smith	L-11
L-15	Perryridge	1500	Smith	L-11
L-14	Downtown	1500	Smith	L-11
L-11	Round Hill	900	Smith	L-11
L-93	Mianus	500	Smith	L-23
L-23	Redwood	2000	Smith	L-23
L-17	Downtown	1000	Smith	L-23
L-16	Perryridge	1300	Smith	L-23
L-15	Perryridge	1500	Smith	L-23
L-14	Downtown	1500	Smith	L-23
L-11	Round Hill	900	Smith	L-23
L-93	Mianus	500	Williams	L-17
L-23	Redwood	2000	Williams	L-17
L-17	Downtown	1000	Williams	L-17
L-16	Perryridge	1300	Williams	L-17
L-15	Perryridge	1500	Williams	L-17
L-14	Downtown	1500	Williams	L-17
L-11	Round Hill	900	Williams	L-17

56 rows in set (0.00 sec)

9.

```
mysql> SELECT *
-> FROM loan_relation
-> INNER JOIN borrower_relation ON loan_relation.loan_number = borrower_relation.`loan-number`;
```

loan_number	branch_name	amount	customer-name	loan-number
L-16	Perryridge	1300	Adams	L-16
L-93	Mianus	500	Curry	L-93
L-15	Perryridge	1500	Hayes	L-15
L-14	Downtown	1500	Jackson	L-14
L-17	Downtown	1000	Jones	L-17
L-11	Round Hill	900	Smith	L-11
L-23	Redwood	2000	Smith	L-23
L-17	Downtown	1000	Williams	L-17

8 rows in set (0.00 sec)

10.

```
mysql> SELECT *
      -> FROM loan_relation
      -> NATURAL RIGHT JOIN borrower_relation;
```

customer-name	loan-number	loan_number	branch_name	amount
Adams	L-16	L-93	Mianus	500
Adams	L-16	L-23	Redwood	2000
Adams	L-16	L-17	Downtown	1000
Adams	L-16	L-16	Perryridge	1300
Adams	L-16	L-15	Perryridge	1500
Adams	L-16	L-14	Downtown	1500
Adams	L-16	L-11	Round Hill	900
Curry	L-93	L-93	Mianus	500
Curry	L-93	L-23	Redwood	2000
Curry	L-93	L-17	Downtown	1000
Curry	L-93	L-16	Perryridge	1300
Curry	L-93	L-15	Perryridge	1500
Curry	L-93	L-14	Downtown	1500
Curry	L-93	L-11	Round Hill	900
Hayes	L-15	L-93	Mianus	500
Hayes	L-15	L-23	Redwood	2000
Hayes	L-15	L-17	Downtown	1000
Hayes	L-15	L-16	Perryridge	1300
Hayes	L-15	L-15	Perryridge	1500
Hayes	L-15	L-14	Downtown	1500
Hayes	L-15	L-11	Round Hill	900
Jackson	L-14	L-93	Mianus	500
Jackson	L-14	L-23	Redwood	2000
Jackson	L-14	L-17	Downtown	1000
Jackson	L-14	L-16	Perryridge	1300
Jackson	L-14	L-15	Perryridge	1500
Jackson	L-14	L-14	Downtown	1500
Jackson	L-14	L-11	Round Hill	900
Jones	L-17	L-93	Mianus	500
Jones	L-17	L-23	Redwood	2000
Jones	L-17	L-17	Downtown	1000
Jones	L-17	L-16	Perryridge	1300
Jones	L-17	L-15	Perryridge	1500

Jackson	L-14	L-93	Mianus	500
Jackson	L-14	L-23	Redwood	2000
Jackson	L-14	L-17	Downtown	1000
Jackson	L-14	L-16	Perryridge	1300
Jackson	L-14	L-15	Perryridge	1500
Jackson	L-14	L-14	Downtown	1500
Jackson	L-14	L-11	Round Hill	900
Jones	L-17	L-93	Mianus	500
Jones	L-17	L-23	Redwood	2000
Jones	L-17	L-17	Downtown	1000
Jones	L-17	L-16	Perryridge	1300
Jones	L-17	L-15	Perryridge	1500
Jones	L-17	L-14	Downtown	1500
Jones	L-17	L-11	Round Hill	900
Smith	L-11	L-93	Mianus	500
Smith	L-11	L-23	Redwood	2000
Smith	L-11	L-17	Downtown	1000
Smith	L-11	L-16	Perryridge	1300
Smith	L-11	L-15	Perryridge	1500
Smith	L-11	L-14	Downtown	1500
Smith	L-11	L-11	Round Hill	900
Smith	L-23	L-93	Mianus	500
Smith	L-23	L-23	Redwood	2000
Smith	L-23	L-17	Downtown	1000
Smith	L-23	L-16	Perryridge	1300
Smith	L-23	L-15	Perryridge	1500
Smith	L-23	L-14	Downtown	1500
Smith	L-23	L-11	Round Hill	900
Williams	L-17	L-93	Mianus	500
Williams	L-17	L-23	Redwood	2000
Williams	L-17	L-17	Downtown	1000
Williams	L-17	L-16	Perryridge	1300
Williams	L-17	L-15	Perryridge	1500
Williams	L-17	L-14	Downtown	1500
Williams	L-17	L-11	Round Hill	900

56 rows in set (0.00 sec)

11.

```
mysql> SELECT *
-> FROM loan_relation
-> RIGHT JOIN borrower_relation ON loan_relation.loan_number = borrower_relation.`loan-number`;
+-----+-----+-----+-----+-----+
| loan_number | branch_name | amount | customer-name | loan-number |
+-----+-----+-----+-----+-----+
| L-16       | Perryridge  | 1300   | Adams         | L-16       |
| L-93       | Mianus      | 500    | Curry         | L-93       |
| L-15       | Perryridge  | 1500   | Hayes         | L-15       |
| L-14       | Downtown   | 1500   | Jackson       | L-14       |
| L-17       | Downtown   | 1000   | Jones         | L-17       |
| L-11       | Round Hill  | 900    | Smith         | L-11       |
| L-23       | Redwood     | 2000   | Smith         | L-23       |
| L-17       | Downtown   | 1000   | Williams      | L-17       |
+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

12.


```
mysql> SELECT *
-> FROM loan_relation
-> NATURAL LEFT JOIN borrower_relation;
```

loan_number	branch_name	amount	customer-name	loan-number
L-11	Round Hill	900	Williams	L-17
L-11	Round Hill	900	Smith	L-23
L-11	Round Hill	900	Smith	L-11
L-11	Round Hill	900	Jones	L-17
L-11	Round Hill	900	Jackson	L-14
L-11	Round Hill	900	Hayes	L-15
L-11	Round Hill	900	Curry	L-93
L-11	Round Hill	900	Adams	L-16
L-14	Downtown	1500	Williams	L-17
L-14	Downtown	1500	Smith	L-23
L-14	Downtown	1500	Smith	L-11
L-14	Downtown	1500	Jones	L-17
L-14	Downtown	1500	Jackson	L-14
L-14	Downtown	1500	Hayes	L-15
L-14	Downtown	1500	Curry	L-93
L-14	Downtown	1500	Adams	L-16
L-15	Perryridge	1500	Williams	L-17
L-15	Perryridge	1500	Smith	L-23
L-15	Perryridge	1500	Smith	L-11
L-15	Perryridge	1500	Jones	L-17
L-15	Perryridge	1500	Jackson	L-14
L-15	Perryridge	1500	Hayes	L-15
L-15	Perryridge	1500	Curry	L-93
L-15	Perryridge	1500	Adams	L-16
L-16	Perryridge	1300	Williams	L-17
L-16	Perryridge	1300	Smith	L-23
L-16	Perryridge	1300	Smith	L-11
L-16	Perryridge	1300	Jones	L-17
L-16	Perryridge	1300	Jackson	L-14
L-16	Perryridge	1300	Hayes	L-15
L-16	Perryridge	1300	Curry	L-93
L-16	Perryridge	1300	Adams	L-16
L-17	Downtown	1000	Williams	L-17

L-16	Perryridge	1300	Jackson	L-14
L-16	Perryridge	1300	Hayes	L-15
L-16	Perryridge	1300	Curry	L-93
L-16	Perryridge	1300	Adams	L-16
L-17	Downtown	1000	Williams	L-17
L-17	Downtown	1000	Smith	L-23
L-17	Downtown	1000	Smith	L-11
L-17	Downtown	1000	Jones	L-17
L-17	Downtown	1000	Jackson	L-14
L-17	Downtown	1000	Hayes	L-15
L-17	Downtown	1000	Curry	L-93
L-17	Downtown	1000	Adams	L-16
L-23	Redwood	2000	Williams	L-17
L-23	Redwood	2000	Smith	L-23
L-23	Redwood	2000	Smith	L-11
L-23	Redwood	2000	Jones	L-17
L-23	Redwood	2000	Jackson	L-14
L-23	Redwood	2000	Hayes	L-15
L-23	Redwood	2000	Curry	L-93
L-23	Redwood	2000	Adams	L-16
L-93	Mianus	500	Williams	L-17
L-93	Mianus	500	Smith	L-23
L-93	Mianus	500	Smith	L-11
L-93	Mianus	500	Jones	L-17
L-93	Mianus	500	Jackson	L-14
L-93	Mianus	500	Hayes	L-15
L-93	Mianus	500	Curry	L-93
L-93	Mianus	500	Adams	L-16

56 rows in set (0.00 sec)

13.

```
mysql> SELECT *
-> FROM loan_relation
-> LEFT JOIN borrower_relation ON loan_relation.loan_number = borrower_relation.`loan-number`;
```

loan_number	branch_name	amount	customer-name	loan-number
L-11	Round Hill	900	Smith	L-11
L-14	Downtown	1500	Jackson	L-14
L-15	Perryridge	1500	Hayes	L-15
L-16	Perryridge	1300	Adams	L-16
L-17	Downtown	1000	Williams	L-17
L-17	Downtown	1000	Jones	L-17
L-23	Redwood	2000	Smith	L-23
L-93	Mianus	500	Curry	L-93

8 rows in set (0.00 sec)

14.

```
mysql> SELECT *
-> FROM loan_relation
-> LEFT JOIN borrower_relation ON loan_relation.loan_number = borrower_relation.`loan-number`
->
-> UNION
->
-> SELECT *
-> FROM loan_relation
-> RIGHT JOIN borrower_relation ON loan_relation.loan_number = borrower_relation.`loan-number`;
```

loan_number	branch_name	amount	customer-name	loan-number
L-11	Round Hill	900	Smith	L-11
L-14	Downtown	1500	Jackson	L-14
L-15	Perryridge	1500	Hayes	L-15
L-16	Perryridge	1300	Adams	L-16
L-17	Downtown	1000	Williams	L-17
L-17	Downtown	1000	Jones	L-17
L-23	Redwood	2000	Smith	L-23
L-93	Mianus	500	Curry	L-93

8 rows in set (0.00 sec)